

AI Agent Planning & Design Document

Data Analyst AI Agent

Date: 24 September 2026

Project: AI Agent Design

Agent Type: Data Analyst Agent

1. Introduction

The **Data Analyst AI Agent** is an AI-powered assistant designed to help employees analyze business data using natural-language questions.

Instead of manually writing SQL queries, Python programs, or creating charts, users can ask questions in simple English. The AI Agent understands the request, decides which tools are required, retrieves and analyzes the data, and presents the results in an easy-to-understand format.

2. Objective

The main objective of the Data Analyst AI Agent is to:

- Allow users to analyze data using natural-language queries.
- Reduce the need for manual SQL queries and data-processing code.
- Automatically select appropriate tools for different analysis tasks.
- Generate meaningful insights from company data.
- Create tables, charts, and summaries when required.
- Help employees make data-driven decisions.

Example

Instead of writing a SQL query such as:

```
SELECT region, SUM(sales)
FROM sales
GROUP BY region;
```

the user can simply ask:

"Show me total sales by region."

The AI Agent handles the technical work and returns the result.

3. Problem Statement

Employees often need technical knowledge of SQL, Python, Excel, or data visualization tools to analyze company data.

This can make simple data analysis time-consuming.

The Data Analyst AI Agent solves this problem by providing a natural-language interface between the employee and the company's data.

Traditional Approach

```
Employee
↓
Write SQL
↓
Execute Query
↓
Export Data
↓
Analyze Data
↓
Create Charts
↓
Prepare Report
```

AI Agent Approach

```
Employee
↓
Ask Question
↓
AI Agent
↓
Data Analysis
↓
Result + Chart + Insights
```

4. Use Case Description

Use Case: Sales Performance Analysis

Scenario

A company employee wants to analyze the previous month's sales performance.

The employee asks:

"Which regions had the highest and lowest sales last month?"

Agent Process

1. The agent receives the user's question.
2. The LLM identifies the user's intent.
3. The agent determines which data is required.
4. The agent selects the SQL/database tool.
5. The SQL tool retrieves the required sales data.
6. Python/Pandas can be used for additional calculations.
7. The agent compares sales across regions.
8. A chart can be generated if required.
9. The LLM summarizes the findings.
10. The final result is presented to the user.

Example Output

Region Sales

North ₹12.5 L
South ₹10.8 L
East ₹8.9 L
West ₹11.2 L

The agent can then provide a short explanation of the results and optionally generate a visualization.

5. Available Tools

The Data Analyst Agent can use multiple tools depending on the user's request.

Tool	Purpose
LLM	Understand requests, planning, reasoning, and response generation
SQL Tool	Generate and execute database queries
Database	Store and provide company data
Python	Perform calculations and advanced analysis
Pandas	Data cleaning, filtering, grouping, and analysis
Matplotlib	Generate charts and visualizations
CSV/Excel Reader	Read uploaded datasets

The agent does **not necessarily use every tool for every request**. It selects tools based on the task.

6. Decision-Making Process

The agent follows a tool-selection process.

Step 1 – Understand the Request

The LLM analyzes the user's natural-language question.

Example:

"How many customers registered this month?"

The agent identifies this as a simple counting operation.

Step 2 – Identify Required Data

The agent determines that customer registration data is required.

Step 3 – Select the Appropriate Tool

The agent selects SQL because the required information is stored in the database.

Step 4 – Execute the Task

The SQL tool retrieves the required information.

Step 5 – Analyze the Result

The agent checks the returned data and performs additional calculations if necessary.

Step 6 – Generate the Response

The LLM converts the result into a simple explanation for the user.

7. Tool Selection Examples

Simple Database Question

User:

"How many customers registered this month?"

```
User
↓
LLM
↓
SQL Tool
↓
Database
```

↓
Result
↓
LLM
↓
Answer

Data Analysis Question

User:

"Calculate the average sales for each region."

User
↓
LLM
↓
SQL Tool
↓
Database
↓
Python / Pandas
↓
Analysis
↓
LLM
↓
Answer

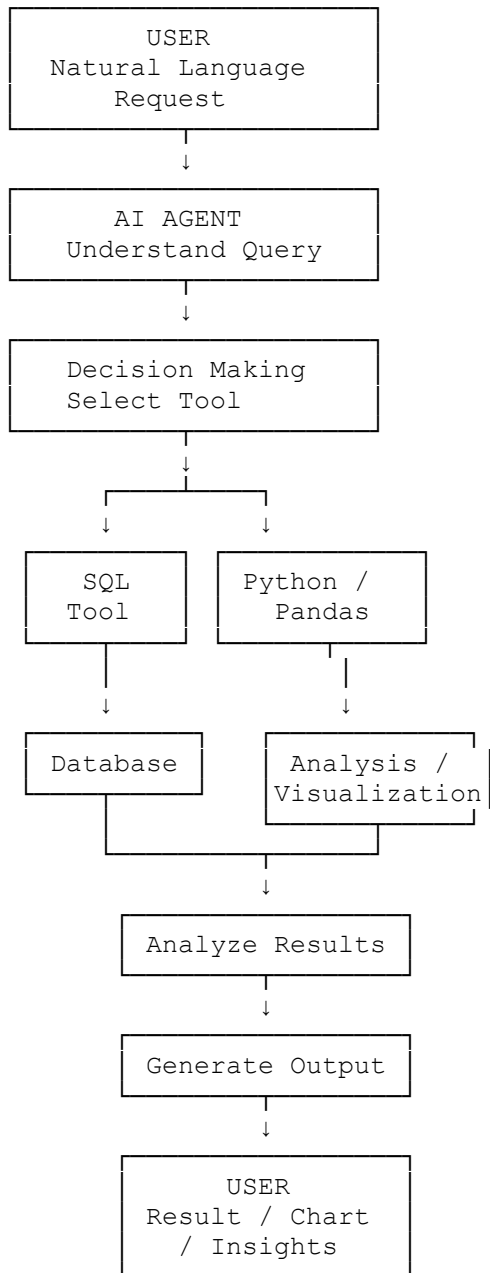
Visualization Request

User:

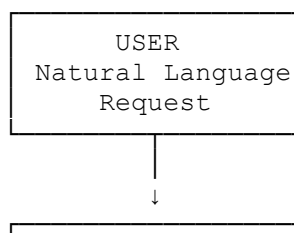
"Show me the sales trend for the last 12 months."

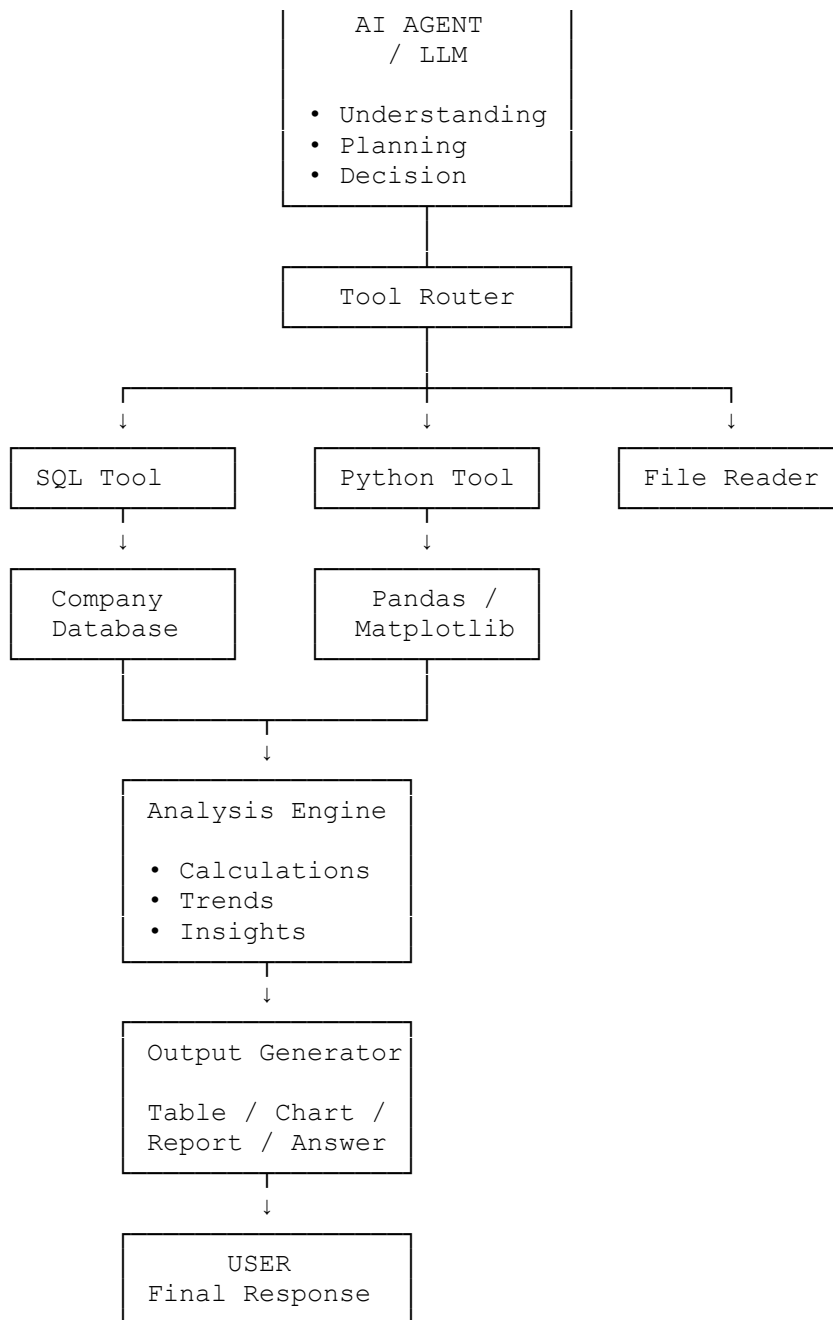
User
↓
LLM
↓
SQL Tool
↓
Database
↓
Python / Pandas
↓
Matplotlib
↓
Chart
↓
LLM
↓
Explanation

8. Task Execution Workflow



9. AI Agent Architecture





10. Expected Outputs

Depending on the user's request, the agent can generate:

Numerical Results

Example:

Total sales this month: ₹45.2 lakh.

Tables

Region	Sales
North	₹12.5L
South	₹10.8L
East	₹8.9L
West	₹11.2L

Charts

- Bar charts
- Line charts
- Pie charts
- Trend charts

Insights

Example:

"Sales increased compared with the previous month, while the East region showed a decline."

Reports

The agent can combine calculations, tables, visualizations, and explanations into a structured report.

11. Example End-to-End Execution

User Request

"Analyze last month's sales and show me the sales trend by region."

Agent Execution

1. Receive request
- ↓
2. Understand user intent
- ↓
3. Identify required sales data
- ↓
4. Select SQL tool
- ↓
5. Retrieve sales data
- ↓
6. Process data using Pandas
- ↓
7. Calculate regional trends
- ↓
8. Generate chart using Matplotlib
- ↓
9. Generate explanation using LLM
- ↓
10. Return chart + analysis to user

12. Benefits

The Data Analyst AI Agent can provide several benefits:

- Reduces manual data analysis.
- Makes data analysis accessible to non-technical employees.
- Saves time when generating SQL queries.
- Automates repetitive analysis tasks.
- Generates visualizations automatically.
- Provides faster access to business insights.
- Allows employees to interact with data using natural language.

13. Future Enhancements

The agent can later be extended with:

- Role-based access control.
- Multiple database connections.
- Automated report generation.
- Scheduled reports.
- Advanced statistical analysis.
- Machine learning predictions.
- Anomaly detection.
- Natural-language SQL generation.
- Conversation history.
- Audit logging.
- Data privacy and security controls.

14. Conclusion

The **Data Analyst AI Agent** provides a natural-language interface for interacting with business data.

The agent understands the user's request, determines the required actions, selects appropriate tools such as SQL, Python, and Pandas, analyzes the retrieved information, and presents the results through tables, charts, insights, or reports.

This architecture can help organizations reduce manual data-analysis work and make business information easier for employees to access and understand.